

From a Single Jump to a Point Mass: A Story of Heaviside and Dirac

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Opening: Why Revisit a “Simple” Function?

It seems we may have overlooked some special properties of very simple functions that we use all the time. In my study of integral transforms, I was struck by a basic observation: a step function with only one jump point already carries unexpectedly rich content in distribution theory and measure theory [1]. This note tells that story, from a jump discontinuity to a point mass.

Act I: The Step Function and Its First Shape

Define

$$H(t - a) = \begin{cases} 0, & t < a, \\ 1, & t > a. \end{cases}$$

This function has a unit jump at $t = a$ and acts like a switch. The value at $t = a$ can be chosen by convention (for instance $H(0) = \frac{1}{2}$), but that choice does not affect the distributional derivative.

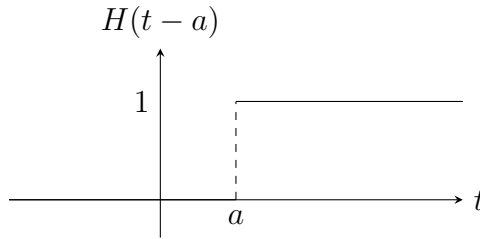


Figure 1: The Heaviside unit step

A rectangle over $[a, b]$ can be written as

$$f(t) = H(t - a) - H(t - b).$$

So even at this stage, a jump already builds geometric structure.

Act II: Squeezing Area into a Point

Consider a rectangular pulse of width 2ε and height $\frac{1}{2\varepsilon}$:

$$d_a(t) = \frac{1}{2\varepsilon} \left[H(t - a + \varepsilon) - H(t - a - \varepsilon) \right],$$

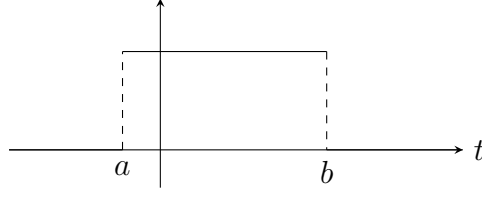


Figure 2: Rectangle function $f(t) = H(t - a) - H(t - b)$

with unit area. Letting $\varepsilon \rightarrow 0$ yields the idealized impulse (Dirac δ):

$$\delta(t - a) = \lim_{\varepsilon \rightarrow 0} d_a(t), \quad \int_{-\infty}^{\infty} \delta(t - a) dt = 1.$$

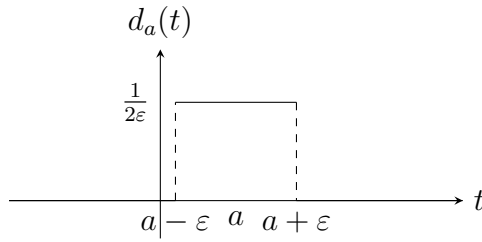


Figure 3: Rectangular approximation of the impulse

Act III: The Distribution Appears

For a suitable test function f ,

$$\int_{-\infty}^{\infty} \delta(t - a) f(t) dt = f(a).$$

This sifting identity shows why δ is not an ordinary function: it is meaningful through its action inside integrals, i.e., as a distribution.

Now the key step in the story: for a test function $\varphi \in C_c^\infty(\mathbb{R})$,

$$\langle (H(t - a))', \varphi \rangle = -\langle H(t - a), \varphi' \rangle = -\int_a^\infty \varphi'(t) dt = \varphi(a) = \langle \delta(t - a), \varphi \rangle.$$

Hence, in the distributional sense,

$$(H(t - a))' = \delta(t - a).$$

So the derivative of the jump is a point impulse.

Act IV: Measure-Theoretic Meaning (Core Viewpoint)

In measure language, for a point a , the Dirac measure δ_a assigns mass 1 to any set containing a and 0 otherwise, so the sifting property becomes

$$\int f(t) d\delta_a(t) = f(a).$$

The Heaviside function is discontinuous, but as a distribution (or equivalently, in the Stieltjes integral sense) its derivative is a measure concentrated at the jump point. Thus the statement $(H(t-a))' = \delta(t-a)$ expresses that the distributional derivative of a jump is a point mass.

Lebesgue measure (context). The Lebesgue measure on $\mathbb{R}$ is the standard measure used to quantify the “size” of sets. For an interval $[a, b]$ it equals the length,

$$\mu([a, b]) = b - a.$$

It is a regular measure on the Borel sets and is countably additive: for any countable collection of measurable sets $\{A_i\}$,

$$\mu\left(\bigcup_i A_i\right) = \sum_i \mu(A_i).$$

Lebesgue measure applies to many kinds of sets (for instance, dense sets and the set of irrationals), but it assigns zero measure to every singleton:

$$\mu(\{a\}) = 0 \quad \text{for all } a \in \mathbb{R}.$$

Dirac vs. Lebesgue (measure viewpoint). The Dirac δ is not an ordinary function; it is a distribution that corresponds to the *Dirac measure* δ_a , a point mass at a . In contrast, Lebesgue measure describes the size of extended sets. A useful comparison:

- **Support.** Lebesgue measure has support $\mathbb{R}$, while δ_a is supported only at the single point $t = a$.
- **Values.** Lebesgue measure gives $\mu(\{a\}) = 0$ and $\mu([a, b]) = b - a$. Dirac measure gives $\delta_a(A) = 1$ if $a \in A$ and 0 otherwise.
- **Additivity and regularity.** Both are genuine measures: μ and δ_a are countably additive and regular on Borel sets. The Dirac measure is “extreme” only in the sense that all of its mass is concentrated at a single point.

Act V: Consequences and Proofs (Preserving the Toolkit)

Consequences and proofs. We use the distributional identity $f(t)\delta(t-a) = f(a)\delta(t-a)$.

- (a) $t\delta(t) = 0$. Let $f(t) = t$. Then $t\delta(t) = f(t)\delta(t) = f(0)\delta(t) = 0 \cdot \delta(t) = 0$.
 (b) $\delta(at) = \frac{1}{|a|}\delta(t)$ for $a \neq 0$. For any test function φ ,

$$\int_{-\infty}^{\infty} \delta(at)\varphi(t) dt = \int_{-\infty}^{\infty} \delta(u)\varphi(u/a) \frac{du}{|a|} = \frac{1}{|a|}\varphi(0) = \int_{-\infty}^{\infty} \frac{1}{|a|}\delta(t)\varphi(t) dt.$$

Thus $\delta(at) = \frac{1}{|a|}\delta(t)$.

- (c) $\delta(t-a) = \delta(a-t)$. Use (b) with $a = -1$:

$$\delta(a-t) = \delta(-(t-a)) = \delta(-1 \cdot (t-a)) = \delta(t-a).$$

(d) If g is monotone, C^1 near a , $g(a) = 0$, and $g'(a) \neq 0$, then $\delta(g(t)) = \frac{1}{|g'(a)|} \delta(t - a)$. For a test function φ , substitute $u = g(t)$ (monotonicity gives a valid change of variables). Then

$$\int \delta(g(t)) \varphi(t) dt = \int \delta(u) \varphi(g^{-1}(u)) \frac{du}{|g'(g^{-1}(u))|} = \frac{\varphi(a)}{|g'(a)|}.$$

Hence $\delta(g(t)) = \frac{1}{|g'(a)|} \delta(t - a)$.

(e) From (d), $\delta(at - b) = \frac{1}{|a|} \delta(t - b/a)$. Let $g(t) = at - b$ so $g(a_0) = 0$ at $a_0 = b/a$ and $g'(t) = a$.

Closing Remark

The central message is simple but deep: in classical calculus, jumps look singular; in measure and distribution language, they become precise objects. The step function does not merely “jump” at a point; its derivative records that jump as a point mass.

References

- [1] Larry C. Andrews and Bhimsen K. Shivamoggi. *Integral Transforms for Engineers*. SPIE—The International Society for Optical Engineering, Bellingham, WA, USA, 1999. SPIE Monograph.